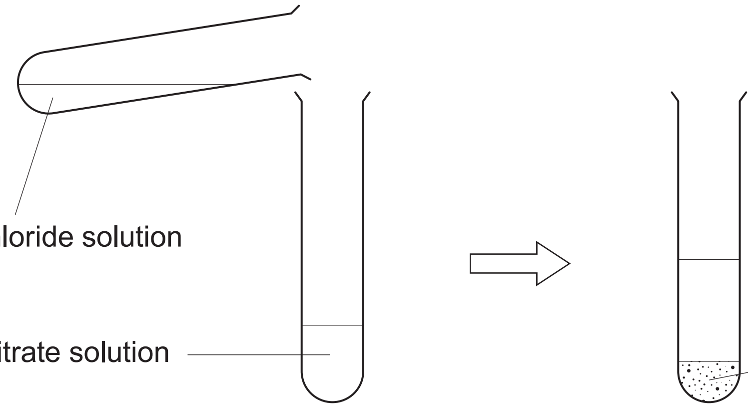
Examiner only

5. *(a)* Dafydd was asked to make some silver chloride. He formed a white precipitate of silver chloride by mixing solutions of sodium chloride and silver nitrate.



(i) Complete the symbol equation for this reaction. [1]

$$NaCl \;+\; AgNO_3 \longrightarrow \ldots\ldots\ldots\ldots \;+\; \ldots\ldots\ldots\ldots$$

(ii) Put a tick (✓) in the box next to the statement which describes why this method works. [1]

silver is more dense than sodium	☐
silver chloride is soluble	☐
silver chloride is insoluble	☐
silver is below sodium in the reactivity series	☐

(iii) Give the name of the process that you would use to separate the precipitate of silver chloride from the reaction mixture. [1]

....

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(b) Calculate the relative formula mass (M_r) of silver nitrate, $AgNO_3$. [2]

$A_r(O) = 16$ $A_r(N) = 14$ $A_r(Ag) = 108$

M_r = ...

(c) The relative formula mass (M_r) of sodium chloride, NaCl, is 58.5.

Calculate the percentage of sodium in sodium chloride. Give your answer to **1 decimal place**. [2]

$A_r(Na) = 23$

Percentage = ... %

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